

BECM 3107

CHANGES TO THE WORK

Term Final Exam Study Guide

Easy English + বাংলা ব্যাখ্যা + Memory Techniques

Rules • Comparisons • Case Study • Vocabulary • MCQs • Model Answers

SOURCE BASIS

Prepared strictly from the uploaded lecture “5.0 - Changes to the Work.” The guide preserves the lecture’s terminology, sequence, examples, stated deadlines, and case study.

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1. Master Map of the Lecture

See the entire chapter as one connected process.

Stage	Main topic	Memory cue
Why	Why changes become necessary	E-S-O-N-R
Plan	CCN evaluates a possible change	CHECK BEFORE COMMIT
Order	Change Order or Change Directive	AGREE vs DIRECT
Price	T&M, lump sum, or third-party determination	COST + O/H + PROFIT
Time	Completion-date adjustment	PRICE AND TIME TOGETHER
Claim	Constructive change and notice	EXTRA WITHOUT FORMAL ORDER
Authority	Who may issue instructions	VERIFY AUTHORITY
Protect	Reservation of rights and records	RESERVE + RECORD
Limit	Cardinal change	MATERIALLY DIFFERENT

Best answer formula: D-R-E-C

D-R-E-C
Definition → Rule → Example → Conclusion. বাংলা: সংজ্ঞা → নিয়ম → উদাহরণ → উপসংহার।

Revision method

- Read one page, close it, and say the mnemonic aloud.
- Write the distinction in a two-column table from memory.
- Practice one 5-mark answer and one 10-mark answer daily.

2. Why Changes Become Necessary

Construction rarely proceeds exactly as originally designed.

Reason	Easy explanation	বাংলা অর্থ
Errors or omissions	Mistakes or missing details in plans/specifications	নকশা বা স্পেসিফিকেশনে ভুল/বাদ পড়া
Unforeseen site conditions	Unexpected physical conditions at site	অপ্রত্যাশিত সাইট অবস্থা
Owner's changing needs	Functional or aesthetic preferences change	মালিকের প্রয়োজন/পছন্দ বদলানো
Cost-saving proposal	Contractor suggests a cheaper method or material	খরচ সাশ্রয়ের প্রস্তাব
New regulations	Government rules require modification	নতুন সরকারি বিধি
Financial setback	Owner revises the budget	আর্থিক সমস্যায় বাজেট সংশোধন

MNEMONIC: E-S-O-C-R-F

Errors/Omissions → Site conditions → Owner needs → Cost savings → Regulations → Financial setback.

Constructive-change warning

Even without an explicit change request, a contractor may believe that it is being required to perform work outside the contract scope. The contractor may then file a constructive-change claim.

BANGLA

Constructive change = আনুষ্ঠানিক change order না থাকলেও বাস্তবে অতিরিক্ত/পরিবর্তিত কাজ করানো।

3. CCN, Change Order and Change Directive

Three different steps - do not confuse them.

Document	When used	Signatures / effect
Contemplated Change Notice (CCN)	Before commitment; owner evaluates possible cost and schedule effects	Evaluation request; not yet the final commitment
Change Order (CO)	Parties have agreed on the change	Written modification signed by owner and contractor; defines scope, price, and schedule effect
Construction Change Directive (CCD)	Work must proceed before price/time agreement	Written order signed by owner, often A/E; price and time negotiated later

MEMORY: THINK - AGREE - DIRECT

CCN = think/evaluate. CO = agree and sign. CCD = direct now, adjust later.

Role of the A/E in a Change Order

- Many contracts require A/E review and signature.
- Purpose: ensure the change does not negatively affect the design.

Necessary words

Word	বাংলা অর্থ
Contemplated	বিবেচনাধীন / সম্ভাব্য
Commitment	চূড়ান্ত অঙ্গীকার
Modification	চুক্তির পরিবর্তন
Directive	নির্দেশ
Scope	কাজের পরিধি

4. Changes Clause and Unilateral Changes

Why an owner may direct a change without prior agreement

General rule

Normally, both parties must agree to the terms of a contract modification.

Exception created by the changes clause

- The changes clause may give the owner the right to make unilateral changes.
- Courts uphold that right because the contractor agreed to it by signing the contract.
- The owner may order additions, deletions, or revisions only within the general scope of the contract.

WITHOUT A CHANGES CLAUSE

A unilateral directive may be a breach of contract, especially where work is deleted.

MEMORY: CLAUSE = CONSENT IN ADVANCE

The contractor's signature is advance agreement that the owner may issue qualifying unilateral changes.

Exam distinction

Situation	Legal effect in the lecture
Valid changes clause + within general scope	Owner's unilateral direction can be enforced
No changes clause	Unilateral direction may be breach
Direction beyond general scope	May become a cardinal change / breach

5. Pricing Change Orders: Time and Materials

Used when the parties cannot agree on a price in advance

T&M RULE

The contractor is reimbursed for costs actually incurred and receives the agreed amount for overhead and profit. T&M resembles a cost-plus contract.

Typical reimbursable items

Cost item	Included example
Labor	Wages, taxes, and benefits
Materials	Materials, supplies, and machinery
Equipment	Rental or use of equipment
Project charges	Bond premiums, insurance, permit fees, or taxes related to the change
Direct supervision	Field office personnel directly attributable to the change

MNEMONIC: L-M-E-B-S

Labor → Materials → Equipment → Bonds/fees → Supervision.

Bangla summary

T&M পদ্ধতিতে পরিবর্তিত কাজের প্রকৃত খরচ ফেরত দেওয়া হয়; তার সঙ্গে নির্ধারিত overhead (পরোক্ষ ব্যয়) ও profit (লাভ) যোগ হয়।

6. Third-Party Price and Schedule Adjustments

When the owner and contractor cannot agree

A. Price determined by a third party

- The contract may authorize a third party, such as the A/E, to determine the change price.
- Risks include favoritism or ignorance.
- Small-price disputes are often not challenged because legal and dispute costs may exceed the possible recovery.

B. Schedule adjustment

- Handled in the same way as price adjustment under many standard forms.
- If agreed, the change order includes a revised completion date.
- If not agreed, a third party such as the A/E may determine the time adjustment.

IMPORTANT LIMITATION

Extra days allowed for the direct changed work may not capture all indirect schedule impacts caused by the change.

MEMORY: PRICE + TIME ARE TWINS

A change can affect both money and completion date; examine both every time.

7. Constructive Changes

A change in reality, even without a formal change order

DEFINITION

A constructive change occurs when the contractor believes it is required to perform work outside the contract scope, but the owner refuses to acknowledge that the work is a change.

Most common source

Errors or ambiguities in plans and specifications.

Contra proferentem and latent ambiguity

- Latent ambiguities are interpreted against the drafter.
- If the contractor adopts a reasonable interpretation, a later owner directive requiring more work can be a constructive change.
- The contractor may then be entitled to additional compensation.

MEMORY: OUTSIDE SCOPE + NO FORMAL ACKNOWLEDGMENT

কাজ চুক্তির বাইরে, কিন্তু মালিক change হিসেবে মানছে না = constructive change claim.

8. Sources of Constructive Changes

How an informal or disputed change arises

Source	Why it may be a constructive change
Errors/ambiguities	Owner's later interpretation demands work beyond a reasonable contract reading
New restrictions	Environmental or inspection restrictions increase effort or cost
Higher standard	Owner demands quality above the stated specification
Improper rejection	Acceptable work is wrongly rejected and must be corrected unnecessarily
Work-method dispute	Owner demands a different method not required by the contract
Measurement dispute	Unit-price quantity or measurement disagreement creates additional work/cost

Where they are most common

- Fixed-price contracts
- Cost-plus contracts with a Guaranteed Maximum Price (GMP)
- They may also arise in unit-price and other contracts.

RECOVERY RULE

When acceptable work is improperly rejected, the contractor may recover the cost of unnecessary corrective work.

9. Notice Requirements for Constructive Changes

Notice before performing the disputed work

CORE RULE

The contractor must give the owner written notice before doing work that it claims is a constructive change.

Why timely notice matters

Owner can...	Practical benefit
Explore cheaper options	Avoid unnecessary high-cost performance
Request cost documentation	Verify labor, materials, and equipment charges
Adjust the budget	Prepare funds for the potential claim
Decide whether to proceed	Control the project before costs accumulate

Effect of failure

Whether lack of notice bars the claim depends on the contract wording and the jurisdiction in which the claim is litigated.

MEMORY: NOTICE BEFORE COST

First notify, then perform and record. বাংলা: আগে লিখিত নোটিশ, পরে কাজ ও রেকর্ড।

10. Waiver of the Notice Requirement

When recovery may still be possible without formal notice

Public contracts

- Courts usually demand strict compliance to protect public funds.
- The rule may be relaxed if the government already knew about the extra work and was not harmed by the missing notice.

Private contracts

- Courts are generally more flexible.
- Recovery may be allowed when the owner directly ordered the work.
- Recovery may be allowed when the owner verbally agreed to pay.
- A history of accepting claims without strict notice may waive the requirement.

Public	Private
Strict notice is the usual rule	More flexible approach
Possible relaxation: actual knowledge + no harm	Direct order, verbal promise, or prior course of conduct may support waiver

MNEMONIC: K-N-H

Government Knew → No prejudice/Harm → possible relaxation.

11. Extra Work versus Additional Work

A distinction sometimes used in notice disputes

EXTRA WORK	ADDITIONAL WORK
• Work completely outside the contract. • বাংলা: চুক্তির সম্পূর্ণ বাইরের কাজ।	• Not expressly stated, but necessary to complete required work. • বাংলা: লেখা নেই, কিন্তু মূল কাজ শেষ করতে প্রয়োজন।

Why the distinction matters

Contractors sometimes use this distinction to avoid strict notice requirements. Some courts allow recovery for essential additional work without proper notice when the owner was not negatively affected.

MEMORY: EXTRA = OUTSIDE; ADDITIONAL = NECESSARY

Extra work বাইরে। Additional work মূল কাজ সম্পূর্ণ করার জন্য প্রয়োজনীয়।

Quick example

Type	Example logic
Extra work	Adding an unrelated facility not included in the project
Additional work	A necessary connecting item not expressly shown but essential to finish the specified system

12. Federal Government Contracts

Private mutual agreement versus government unilateral order

Point	Private construction	Federal government contract
Change order	Mutual agreement between owner and contractor	Unilateral order issued through the Contracting Officer
Prior agreement	Usually defines work, cost, and schedule together	Contractor must comply even without prior agreement
Authority	Owner/authorized project representative	Contracting Officer or properly authorized official

Changes listed in the lecture

- Specifications, drawings, or designs
- How the work is performed
- Government-provided resources or site conditions
- Acceleration of the work

LECTURE REFERENCE

The slide cites "FAR §52.243-49." This guide preserves that citation exactly as presented in the lecture.

MEMORY: S-M-R-A

Specifications → Method → Resources/site → Acceleration.

13. Equitable Adjustments

Fair written adjustment to price and/or time

RULE

When a change increases or decreases the cost or time required, the Contracting Officer must make an equitable adjustment and modify the contract in writing.

Contractor's submission

- Submit a written statement requesting an adjustment.
- State the proposed amount of the price/time adjustment.
- Submit within 30 days, as stated in the lecture.

Preferred timing and supporting documents

- The amount can be determined after work, but should preferably be agreed before work begins.
- Support the proposal with subcontractor or supplier pricing and other backup records.

MEMORY: 30 - WRITE - SUPPORT

Within 30 days → written proposal → backup documents.

Bangla vocabulary

Term	বাংলা অর্থ
Equitable adjustment	ন্যায্য মূল্য/সময় সমন্বয়
Contracting Officer	চুক্তি কর্মকর্তা
Proposed amount	প্রস্তাবিত পরিমাণ
Backup documentation	সমর্থনকারী নথি

14. Escrow of Bid Documents

Keeping bid calculations securely for later evaluation

What may be deposited

- Takeoffs
- Cost data
- Other bid-working documents

Why owners require escrow

- To evaluate change orders.
- To determine fair adjustments.
- To compare the original bid assumptions with claimed change costs.

Where it is common

- Government projects
- Large projects
- Often only the lowest bidder must submit after bid opening.

CONTRACTOR CONCERN

Confidentiality: contractors may dislike disclosing internal estimating information.

MEMORY: LOCK THE BID, CHECK THE CHANGE

Escrow preserves original bid evidence so later change pricing can be tested.

15. Authority to Issue Changes

Never assume every project representative can change the contract

Authority type	Meaning	Example from lecture logic
Express authority	Clearly assigned power	Contracting Officer formally authorized to order changes
Implied authority	Power necessary to perform assigned duties	Representative may reject defective material
Apparent authority	Owner's conduct makes authority reasonably appear to exist	May be relied on in private construction
Ratification	Unauthorized act later treated as approved	Government knows, benefits, and does not stop the work

Private versus government rule

- Private construction: apparent authority may protect a reasonable contractor.
- Government contract: apparent authority is not recognized; actual authority is required.
- Without actual authority, the contractor may not be paid for extra work.

RATIFICATION TEST: KNOW - BENEFIT - DO NOT STOP

Government knows of the unauthorized work, receives the benefit, and allows it to continue.

16. Duty to Perform Changed Work

Perform now, resolve compensation later - but only within limits

General duty

When the owner requests a change under the contract, the contractor may have to perform without first knowing how much - or even whether - it will be compensated.

Limit on the owner's power

- The owner's right to require changed or extra work is not unlimited.
- If a court finds the work outside the contract scope, the contractor cannot be terminated for refusing that directive.

MEMORY: PERFORM WITHIN SCOPE

Within scope: duty to proceed may apply. Outside scope: refusal may be justified.

Risk warning

The scope question can be difficult. A contractor that refuses should recognize the risk that a later court may find the directive valid.

Within general scope	Outside general scope
Proceed and preserve the price/time claim	Directive may be breach or cardinal change
Follow notice and documentation rules	Contractor may have a right to refuse

17. Reservation of Rights

Protecting claims when the full impact is not yet known

Accord and satisfaction risk

- A change order may say the payment is full and final.
- The release may cover direct costs, indirect costs, and delay impacts.
- Such language can block later claims.

Reservation of rights

- Clearly states that unknown or future impacts are not waived.
- Allows later claims for delays or unforeseen consequences.
- Courts generally support a clearly stated reservation.

MEMORY: SIGN, BUT RESERVE

Agree to the immediate change while expressly preserving claims whose full cost/time impact is not yet known.

Term	Bangla meaning
Accord and satisfaction	পূর্ণ ও চূড়ান্ত নিষ্পত্তি
Release of claims	দাবি ত্যাগ/মুক্তির ধারা
Reservation of rights	অধিকার সংরক্ষণ
Indirect cost	পরোক্ষ ব্যয়
Unforeseen impact	অপ্রত্যাশিত প্রভাব

18. Case Study: Bechtel National, Inc. v. Roche

Accord and satisfaction in a change order

Step	Lecture facts
1. Problem	Errors and ambiguities in bid documents caused delays and increased costs on underground storage facilities.
2. Settlement	The parties signed a \$535,000 change order containing a release of all claims related to the issues.
3. Later claim	The contractor later sought about \$4.7 million more, arguing that the first payment covered only direct construction cost.
4. Decision	The court treated the \$535,000 change order as complete and final settlement; no additional compensation was allowed.

CASE LESSON

A broad release can extinguish later delay and indirect-cost claims. A contractor should insert a clear reservation of rights before signing when full impacts remain unknown.

MEMORY: \$535K RELEASE DEFEATED \$4.7M LATER CLAIM

Release language controlled the result.

19. Changes Clauses in Subcontracts

How change risk flows down the contractual chain

Main principles

- Subcontractors perform much of the actual construction and are heavily affected by changes.
- Subcontracts usually mirror the prime contract, so subcontractors must follow change directives.
- They may also have to perform constructive changes and seek payment afterward.
- They generally recover through the general contractor.

Payment clauses

Clause	Lecture explanation
Pay-if-paid	Subcontractor payment is tied to whether the general contractor receives payment from the owner.
Pay-when-paid	Payment timing is tied to the general contractor's receipt of payment.

MEMORY: OWNER → GC → SUB

Change instructions and payment claims commonly flow through the general contractor.

Bangla

Prime contract-এর পরিবর্তনের নিয়ম সাধারণত subcontract-এ “flow down” করে; অর্থাৎ subcontractor-কেও নির্দেশ মানতে ও GC-এর মাধ্যমে দাবি করতে হয়।

20. Documentation of Costs

A valid change claim needs reliable evidence

Records to maintain

- Daily or weekly time records
- Labor records and payroll support
- Materials and equipment records
- Invoices, supplier quotations, and subcontractor pricing
- Records separating change costs from original-contract work

Consequences of poor records

- The contractor may lose the right to compensation.
- Courts prefer documented actual costs over estimates when tracking was possible.
- Government equitable adjustments are based on reasonable actual costs.

MNEMONIC: D-A-S

Daily records → Actual costs → Separate changed work.

EXAM SENTENCE

Good documentation proves causation and quantum: what the change caused and how much it actually cost.

21. Cardinal Changes

A change so large that it becomes a different undertaking

DEFINITION

A cardinal change goes so far beyond the original contract scope that the project becomes a materially different undertaking. Imposing it amounts to breach of contract.

How it may arise

- One isolated but extremely significant change.
- A series of related or unrelated changes whose cumulative effect transforms the project.

Contractor's options

- May refuse the work and treat the contract as terminated.
- This option is used only in extreme cases because it is risky.
- If a court later finds there was no cardinal change, the contractor may be held in default for unjustified refusal/termination.

MEMORY: CARDINAL = CONTRACT TRANSFORMED

Ordinary change modifies the project; cardinal change materially replaces the undertaking.

22. High-Value Comparison Tables

Use these tables for 5-mark and 10-mark questions.

Point	Change Order	Construction Change Directive	Constructive Change
Form	Written mutual modification	Written unilateral direction under contract	No formal change order acknowledged
Agreement	Owner and contractor agree	Price/time may be unresolved	Owner disputes that a change exists
Timing	Usually before changed work	Work proceeds; adjustment later	Contractor claims after/in connection with extra work
Main issue	Scope, price, schedule settled	Authority and later adjustment	Scope, notice, proof, compensation

Constructive versus cardinal change

Point	Constructive change	Cardinal change
Scope	Extra requirement disputed but claimable under changes process	So far beyond scope that project is materially different
Typical remedy	Additional compensation/time adjustment	Breach remedies; possible refusal/termination
Contractor approach	Give notice, perform, document, claim	Extreme caution; refusal carries default risk

Private versus government authority

Private	Government
Apparent authority may be recognized	Actual express/implied authority required
Courts may flex notice requirements more readily	Strict compliance commonly protects public funds

23. Essential Vocabulary and Memory Codes

Words that should appear correctly in exam answers

English term	বাংলা অর্থ
Change order	পরিবর্তন আদেশ / লিখিত চুক্তি সংশোধন
Construction change directive	নির্মাণ পরিবর্তন নির্দেশ
Constructive change	আনুষ্ঠানিক আদেশ ছাড়া কার্যত পরিবর্তন
Unilateral	একতরফা
Time and materials	সময় ও উপকরণভিত্তিক মূল্য
Overhead	পরোক্ষ/প্রশাসনিক ব্যয়
Equitable adjustment	ন্যায্য মূল্য/সময় সমন্বয়
Escrow	নিরাপদ হেফাজতে জমা
Actual authority	প্রকৃত অনুমোদিত ক্ষমতা
Apparent authority	দৃষ্টিগোচর/প্রতীয়মান ক্ষমতা
Ratification	পরে অনুমোদন/সমর্থন
Reservation of rights	অধিকার সংরক্ষণ
Accord and satisfaction	পূর্ণ ও চূড়ান্ত নিষ্পত্তি
Cardinal change	মূল চুক্তিকে আমূল বদলে দেওয়া পরিবর্তন

Memory-code bank

Code	Meaning
THINK-AGREE-DIRECT	CCN → Change Order → Change Directive
L-M-E-B-S	T&M cost heads
NOTICE BEFORE COST	Constructive-change procedure
30-WRITE-SUPPORT	Equitable adjustment submission
KNOW-BENEFIT-NOT STOP	Ratification
RESERVE + RECORD	Protect future claims
MATERIALLY DIFFERENT	Cardinal change test

24. Model Written Answers

Compact answers you can expand in the exam

Q1. Distinguish Change Order and Construction Change Directive.

A change order is a written mutual modification signed by the owner and contractor that defines the changed scope, price, and schedule effect. A construction change directive is a written owner-issued order, often signed by the A/E, requiring work to proceed before the parties have fully agreed on price or time. The adjustment is negotiated later under the contract. Therefore, the main distinction is prior mutual agreement versus unilateral direction with later adjustment.

Q2. What is a constructive change?

A constructive change arises when a contractor is required to perform work outside the contract scope but the owner does not formally recognize the requirement as a change. Common sources are ambiguous documents, higher standards, new restrictions, and improper rejection of acceptable work. The contractor should give written notice, perform as required within the contract process, keep separate records, and claim additional price or time.

Q3. Explain reservation of rights.

A reservation of rights is a clear statement in a change order preserving the contractor's ability to claim future costs or delay impacts that are unknown when the order is signed. Without it, full-and-final release language may create accord and satisfaction and bar later claims.

Q4. Define cardinal change.

A cardinal change is a change or series of changes so far beyond the original scope that the project becomes a materially different undertaking. It amounts to breach. The contractor may refuse and treat the contract as terminated, but risks default if a court later rejects the cardinal-change argument.

25. MCQ Practice

Answer before looking at the key.

1. A CCN is mainly used to:

- A. Terminate the contractor B. Evaluate a possible change before commitment C. Approve final payment D. Waive notice

2. A signed mutual modification defining scope, price, and schedule is a:

- A. Change order B. Ratification C. Cardinal change D. Escrow

3. T&M pricing usually reimburses costs plus:

- A. Liquidated damages B. Overhead and profit C. Bid security D. Retention only

4. The most common source of constructive changes is:

- A. Insurance B. Errors or ambiguities C. Completion certificate D. Arbitration

5. Before claimed constructive-change work, the contractor should normally give:

- A. Oral praise B. Written notice C. Final release D. Bid bond

6. In government contracts, apparent authority is:

- A. Always enough B. Generally not recognized C. Stronger than express authority D. The same as ratification

7. The lecture states equitable-adjustment requests should be submitted within:

- A. 7 days B. 14 days C. 30 days D. 90 days

8. A reservation of rights protects:

- A. Unknown future claims B. The owner's design copyright C. Bid validity D. Insurance premiums

9. Courts prefer change-cost claims supported by:

- A. Guesses B. Actual documented records C. Oral memory only D. No separation of costs

10. A materially different undertaking is a:

- A. Minor change B. Cardinal change C. Submittal D. CCN

ANSWER KEY

1-B, 2-A, 3-B, 4-B, 5-B, 6-B, 7-C, 8-A, 9-B, 10-B

26. Probable Questions and Final Revision

Use this as your last-day checklist.

Probable written questions

- Discuss the reasons why changes to construction work may be necessary.
- Distinguish CCN, Change Order, and Construction Change Directive.
- Explain T&M pricing and the recoverable cost heads.
- Define constructive change and discuss its common sources.
- Explain notice requirements and waiver in public and private contracts.
- Distinguish extra work from additional work.
- Explain authority to issue changes, apparent authority, and ratification.
- Discuss equitable adjustments and escrow of bid documents.
- Explain reservation of rights with the Bechtel case study.
- Discuss documentation of costs and cardinal changes.

One-minute final recall

FLOW

Reason → CCN → CO/CCD → Price + Time → Notice → Authority → Perform → Reserve → Record → Cardinal limit.

Four-day revision cycle

Day	Task
Day 1	Read explanations and mark Bangla vocabulary.
Day 2	Recall all mnemonics without looking.
Day 3	Write four model answers under time.
Day 4	Solve MCQs and revise comparison tables.

FINAL EXAM RULE

Always mention scope, authority, notice, price/time adjustment, and documentation when analysing a change dispute.